



Instructor: Date:

Learner: Regional Branch:

Program: Center:

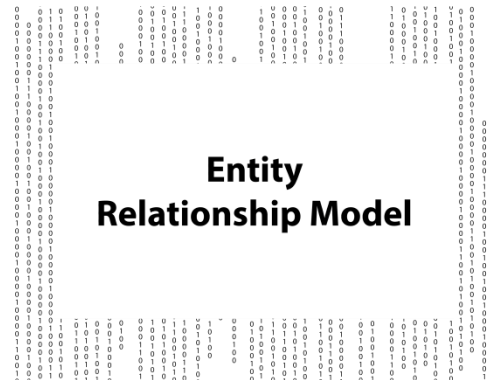
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Understanding Different Types of Relationships



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.

Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can recall information about the ER model in DBMS.
- I can show understanding of basic concepts in the ER model in DBMS.
- I can talk about different kind of relationships in the ER model.

Ask your partner:

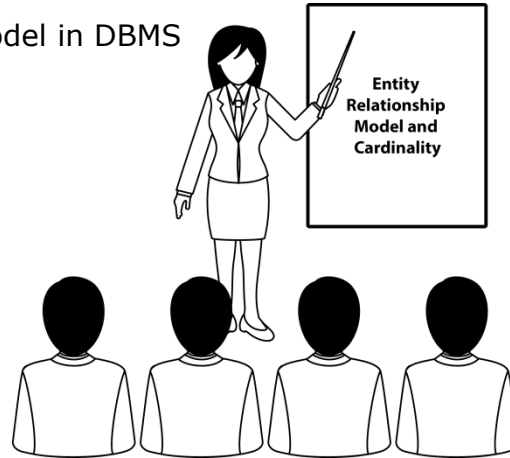
- What types of relationships do you know in computing? Explain yourself with some examples.
- How do relationships in computing work? Describe.





Let's Explore!

A lecture on the ER model in DBMS



Instructions: Read the text below.

The ER model for the database management system can be complicated and difficult to understand. Today you can count on me to clarify all your doubts and questions regarding this topic.

The ER model is based on some simple elements you should have very clear. The first element to check is the entity. An entity is a piece of information that can be easily distinguishable, it can be part or not of the real world. Then, we have the attribute. The attributes are the characteristics that define each entity.

There are several kinds of attributes: the simple attribute, which are atomic values that cannot be divided further; the composite attribute, which are made of more than one attribute; the derived attribute, which are attributes that do not exist in the physical database but are derived from other attributes; the single value attribute, which contain a single value; and the multi-value attribute, which may contain more than one attributes. These kinds of attributes make each reference to the different ways values can be part of the attributes.

Next, you need to watch out for the entity sets and the keys that work together. An entity set is a group of related types of entities, and a key is an attribute that distinguishes one particular (unique) entity from the set. There are three types of keys: the superkey, a candidate key, and a primary key.

Without the ER model, the DBMS could not work, and it is here when the relationships become important. A relationship is a way two entities are related. There are three types of relationships the binary, the ternary and the n-ary. As you can see the number of participants in the relationships is what defines the type of relationship.

To wrap things up, I would like to go very quickly over the cardinalities, which are the different way that entity sets are associated. Cardinalities can be one to one, one to many, many to one or many to many.



**Let's Practice!****Some Facts About the ER Model**

Instructions: Read the following sentences based on the text. Circle True or False.

1. Relationships can be defined as the way two entities are related. ()
2. In a college database, learners, instructors, classes, and courses offered can be considered as entities. ()
3. For the kinds of relationships just two can be included: binary and ternary. ()
4. A key cannot be thought as an attribute that distinguishes one particular or single entity. ()
5. An attribute can be referred to as the synonym for characteristic or distinguishing feature. ()

**Let's Practice!****Mapping Cardinalities**

Instructions: Read the paragraph. Write the missing word(s) in the blank. Use: One to one, ternary, Many to many, binary, One to many, n-nary, Many to one

There are some types of relationships. To define such types the number of participants or entities need to be taken into account as they can be _____, _____ and _____. Once the relationships have been defined into entity sets, the cardinality mapping classifies such sets and the entities alone based on how they relate to each other: _____: an entity from one set related to one entity in another set. _____: one entity from one set related to more than one entity in another set. _____: more than one entity from one set related to only one entity in another entity set. _____: one entity from an entity set related to many entities in another entity set and vice versa.



Extension Activity

Learning More About ER and Mapping Cardinalities

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information on the ER model and Mapping Cardinalities.
- Create a poster in which you include the main characteristics of ER and Mapping Cardinalities.
- Share with your partners the poster you created.

Self-Assessment

Yes

No

- I look for information on ER and Mapping Cardinalities.
- I create a poster with the main characteristics of ER and Mapping Cardinalities.
- I share with my partners the poster I create.

☐
☐
☐
☐
☐
☐


Target Words

- database management system (DMS)** (noun): a computer software application used to capture and analyze data through the interaction with the user, other applications, and the database itself
e.g. The ER model for the database management system can be complicated and difficult to understand.
- atomic value** (noun): value that cannot be divided.
e.g. The simple attribute, which are atomic values that cannot be divided further.
- derive** (verb): to trace from a source or origin
e.g. The derived attribute, which are attributes that do not exist in the physical database but are derived from other attributes.
- superkey** (noun): combination of columns that identifies in a unique way any row within a relational database management system table
e.g. There are three types of keys: the superkey, a candidate key, and a primary key.
- candidate key** (noun): column, or group of columns, in a table that can uniquely identify any database record without referring to any other data
e.g. There are three types of keys: the super key, a candidate key, and a primary key.



Learning Points

Inseparable phrasal verbs, watch out for, count on

A **phrasal verb** is the combination of a verb with a preposition or adverb (or both) that has a meaning that is different from each of the words that are part of the verb. In other words, its meaning is hard to get at a first glance.

There are two types of phrasal verbs separable and **inseparable phrasal verbs**. Separable phrasal verbs can be separated by other words, while **inseparable phrasal verbs** cannot be separated by other words.

Inseparable phrasal verbs can be transitive, which means they can take a direct object, but you cannot place that direct object in the middle of the phrasal verb. In other words, they cannot be split, That is where their name come from. Here we are going to take a look at two of this kind of verbs:

- **Watch out for (something or someone):** to be careful or cautious
Watch out for the viruses on that web page.
- **Count on (something or someone):** to rely or depend on
*You can **count on** me to make this database model.*

Examples

- *Next you need to watch out for the entity sets and the keys that work together*
- *Today you can count on me to clarify al your doubts and questions on the topic*

Mapping Cardinalities

Cardinality is the relationship between one data table and another.

To **map cardinality** you need to define the kind of relationship existing between entities in given sets.

The cardinalities can be:

- **One to one:** an entity or entities from a set can be linked with other entity in another set.
- **One to many:** one entity from an entity set can be associated to more than one entity in another entity set.
- **Many to one:** more than one entity from an entity set can be related to only one entity in another entity set.
- **Many to many:** one entity from an entity set can be related to many entities in another entity set and vice versa.

Let's look at the image:

